$$F(x)=\begin{cases}0, \text{ if } x<0\\ \frac{x}{8}, \text{if } 0\le x<1\\ \frac{1}{4}+\frac{x}{8}, \text{if } 1\le x<2\\ \frac{3}{4}+\frac{x}{12}, \text{if } 2\le x<3\\ 1, \text{for } 3\le x.\end{cases}$$

(a) Compute: (i) $P(1\le X\le 2)$ and (ii) $P(X=3)$.

(b) Is X a discrete random variable? Justify your answer.

3. The p.d.f. of X is defined as

$$f(x)=\begin{cases}k, & \text{for } 0<x\le 4\\ 0, & \text{otherwise}\end{cases}$$

Find the value of k and also find $P(2\le X\le 4)$.

4. The probability distribution function of a discrete random variable X is

$$f(x)=\begin{cases}2k, & x=1\\ 3k, & x=3\\ 4k, & x=5\\ 0, & \text{otherwise}\end{cases}$$

where k is some constant. Find (a) k and (b) $P(X>2)$.

5. The probability density function of a continuous random variable X is

$$f(x)=\begin{cases}a+bx^2, 0\le x\le 1;\\ 0, \text{ otherwise.}\end{cases}$$

where a and b are some constants. Find (i) a and b if $E(X)=\frac{3}{5}$ (ii) $Var(X)$.

6. Prove that if $E(X)=0$, then $V(X)=E(X^2)$.

7. What is the expected value of a game that works as follows: I flip a coin and, if tails you pay ₹ 2; if heads you pay ₹ 1. In either case I also pay you ₹ 0.50

8. Prove that, (i) $V(aX)=a^2V(X)$ and (ii) $V(X+b)=V(X)$

9. Consider a random variable X with p.d.f

$$f(x)=\begin{cases}3x^2, \text{ if } 0<x<1\\ 0, \text{otherwise}\end{cases}$$

Find $E(X)$ and $V(3X-2)$.

10. The time to failure in thousands of hours of an important piece of electronic equipment used in a manufactured DVD player has the density function

$$f(x)=\begin{cases}2e^{-2x}, & x>0\\ 0, & \text{otherwise}\end{cases}$$

Find the expected life of this piece of equipment.

Summary

- A variable which can assume finite number of possible values or an infinite sequence of countable real numbers is called a discrete random variable.
- Probability mass function (p.m.f.)

$$P_X(x)=p(x)=\begin{cases}P(X=x_i)=p_i=p(x_i) & \text{if } x=x_i, i=1,2,...,n,...\\ 0 & \text{if } x\ne x_i\end{cases}$$

Conditions:

- $p(x_i)\ge 0\ \forall\ i$ and
- $\sum_{i=1}^{\infty} p(x_i)=1$

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